BETTY BLANKSON

+233 551 306554 | blankson.amabetty@gmail.com | linkedin.com/in/bettyblankson/ | https://github.com/bettyblankson

PROFILE SUMMARY

Intelligent systems are only as good as the people who understand them end to end. A Master's student in Intelligent Computing Systems with experience spanning machine learning, embedded systems, software engineering, and AI research and driven by a belief that the best technologists are the ones who can build the thing and question whether it should exist.

EDUCATION

Ashesi University — **Berekuso, Eastern Region**
MSc. Intelligent Computing Systems — **Expected Date of Graduation: June 2027**
Cumulative GPA: 3.78/4.00

Relevant Courses: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Internet of Things, AI and Security, Agile Software Engineering Methods

Ashesi University — **Berekuso, Eastern Region**
BSc. Computer Engineering — **Date of Graduation: June 2022**
Relevant Courses: Embedded systems, Internet of Things, Control Systems Design, Electronics and Circuit Design, Digital Systems and Design, Operating systems

CERTIFICATIONS/AWARDS

Fully funded scholarship (Ashesi University), Masters Education	**August 2025 – June 2027**
Machine Learning and Data Analyst (Blossom Academy)	**August 2024- November 2024**
Leadership and Management Skills (Blossom Academy)	**August 2024- November 2024**
Fully funded scholarship (Mastercard Foundation), Data Analyst Certification	**August 2024- November 2024**
Dean's List, Ashesi University	**December 2019**
Fully funded scholarship (Ashesi University), Undergraduate Education	**August 2017 – June 2020**

WORK EXPERIENCE

Oze Ghana - Osu, Accra, Ghana — **June 2026 to August 2026**
Product & Data Intern
Multilingual Transactional WhatsApp Agent | n8n, Supabase, GPT-4o mini, Meta API

- Built an automated WhatsApp chatbot via the Meta Business API and n8n to ingest informal conversational text for bookkeeping.
- Integrated a multimodal vision pipeline into the bot using GPT-4o mini to parse OCR text, prices, and line items from receipt photos.
- Engineered zero-shot extraction schemas across English, Twi, and French, implementing clarification logic to separate services from inventory.
- Designed modular system prompt architectures and state clarification workflows to handle edge cases, false inputs, and dynamic categorization across products and service catalogs.
- Synchronized real-time database transactions to Supabase and external reporting sheets, establishing persistent records with custom date-mapping schemas.

Oze Ghana - Osu, Accra, Ghana — **October 2024 to April 2025**
Data Analyst

- I analyzed customer data to inform business decisions and drive revenue growth.
- I developed and maintained dashboards and reports to track key performance indicators (KPIs) and provide actionable insights using PowerBI and Redash.
- I conducted data-driven research to identify trends, opportunities, and challenges, and presented findings to stakeholders.

Ashesi University- Berekuso, Eastern Region, Ghana — **August 2022–August 2024**
Teaching Assistant (Linear Algebra, Networks and Data Communication)

- I provided administrative support which included assisting professors with tasks such as managing digital class files, grading and recording scripts.
- I supported and guided students and overall fostered a positive learning environment while maintaining a high level of professionalism and discretion.

RESEARCH AND PROJECTS

Do Safety-Aligned Vision-Language Models Degrade Differently Under Common Image Corruptions? (ML Research - Computer Vision)

Collaborated in a team of 5 to conduct the first controlled evaluation of alignment training's effect on corruption robustness. Accepted at the Generative Computer Vision (GCV) Workshop, IEEE/CVF Computer Vision and Pattern Recognition (CVPR), 2026.

- Evaluated four matched base/aligned VLM pairs (InternVL2.5, Qwen2-VL, Gemma-3-4B, Gemma-3-7B) under the ImageNet-C benchmark across GQA, VQAv2, TextVQA, and MMBench, finding that alignment training neither uniformly improves nor harms robustness.
- Documented a pre-training confound affecting several published VLM robustness studies, showing that comparing instruction-tuned models against raw pretrained checkpoints without SFT misattributes instruction-following gains to robustness.

Standard Takers or Standard Makers? The EU AI Act, African AI Developers, and the Politics of Regulatory Transplantation (AI Policy · Technology Ethics)

Collaborated in a team of 2 to author a peer-reviewed research paper accepted for publication at the Future Technologies Conference (FTC) 2026, Berlin — published in Springer's Lecture Notes in Networks and Systems and indexed in SCOPUS and Web of Science.

- Analysed how the EU AI Act's extraterritorial reach (Article 2(1)(c)) creates compliance obligations for African developers without providing access to the institutional support available to EU-based firms.
- Developed a three-profile developer exposure typology and an adoption framework assessing which elements of the EU AI Act are portable, which require contextual adaptation, and which require independent African construction.
- We reframed the EU Act's high-risk category across five domains showing that risk calculus changes fundamentally in contexts where the alternative to AI is an absent human system.

Campus IoT Environmental Monitoring Network (Embedded Systems · Wireless Networking)

Collaborated in a team of 3 to design and deploy a multi-node IoT sensor network across Ashesi University campus.

- I engineered a heterogeneous wireless network spanning 4 field nodes using dual radio protocols — LoRa 433 MHz for long range and nRF24L01 for short range with a multi-hop relay topology to extend coverage beyond direct radio range.
- I configured an ESP32 gateway to manage two radios on a shared SPI bus, coordinating LoRa and nRF24 listen windows and delivering LED commands to each node immediately after it transmitted.
- I collaborated with the team to build a full data pipeline (Mosquitto MQTT → Python subscriber → PostgreSQL → Grafana) with a browser-based web UI for real-time sensor visualization and remote LED actuation across all nodes.

AI-Powered Backend Generation Platform (Natural Language Processing - Agentic Systems)

Collaborated in a team of 4 to build a multi-agent AI platform automating full backend API generation from natural language to deployed infrastructure.

- Built a hybrid multi-agent system generating dockerized FastAPI + PostgreSQL backends from natural language, orchestrating 6 specialist agents across 4 automated pipeline phases.
- Designed a full-stack application (FastAPI, Next.js, WebSockets, Docker Compose) with a GitHub Actions CI/CD pipeline deployed on DigitalOcean.

End-to-End Student Success Prediction & Progression Modelling (Machine Learning)

Applied unsupervised learning and predictive modeling to real institutional data to uncover actionable insights on student academic trajectories and socioeconomic influences on performance.

- I applied supervised and unsupervised ML techniques (K-means, DBSCAN, GMM) to a large dataset of student records using Python and scikit-learn, segmenting students into distinct performance clusters to help institutions identify at-risk students early and allocate academic support more effectively.
- I built a progression model that provides institutions with a data-driven basis for early intervention, enabling advisors to flag struggling students before critical deadlines and potentially improving retention and graduation outcomes.

CO-CURRICULAR ACTIVITIES

Academic Mentor, Ashesi University **December 2022 – August 2024**

- Coached students who were on probation due to low GPAs outside their classroom lessons in courses such as Linear Algebra, Statistics and Intro to Modelling and Simulation.
- Developed and delivered appropriate intervention strategies, working with students to ensure that they make rapid and sustained progress.

Volunteer, IT consultant and software development lead, Helt App **Nov 2014 – Aug 2015**

- I volunteered to lead a team of three software developers to build a health ecosystem consisting of health consultation, heath practitioner job portal and an ecommerce site for health equipment.
- I Increased project clarity by {80%} by modeling software requirements and business model with the company founder.

Volunteer, Project Management team, Loozeele Initiative **Nov 2013 – Aug 2014**

- Part of planning and execution team for an entrepreneurship program for women who carry load in local markets to make ends meet (Kayayei women) and raised funds from sponsors for startup ideas that were devised by participants of the entrepreneurship program.
- Designed and managed an ecommerce site for products produced by participants of the program.

TECHNICAL SKILLS

- Programming Languages and libraries: Python, C/C++, Java, R, SQL, HTML, CSS, Assembly, Pandas, Seaborn, NumPy, SymPy, Matplotlib
- Software/Tools: Excel, PowerBI, Redash, SOLIDWORKS, MATLAB, AutoCAD, LabVIEW, Simulink, Arduino, PLC Programming, LTSpice, Cisco Packet Tracer, Wireshark.

REFERENCES

Available upon request